

Question 1

3 marks

101

Gina correctly started to answer the following question. Complete her solution.

Question: Solve the following equation for all real values of θ .
Express your answer in radians correct to 3 decimal places.

$$3 \sin^2 \theta - 14 \sin \theta - 5 = 0$$

Gina's solution: $3 \sin^2 \theta - 14 \sin \theta - 5 = 0$
 $(3 \sin \theta + 1)(\sin \theta - 5) = 0$

Question 2

3 marks

102

Find and simplify the 6th term in the binomial expansion of $\left(3x^4 - \frac{1}{x^3}\right)^9$.

Question 3

3 marks

103

The number of times a website is visited can be modeled by the function:

$$A = 800(e)^{rt}$$

where A = the total number of visitors at time t

t = the time in days ($t \geq 0$)

r = the rate of growth

After 5 days, 40 000 people have visited the site.

Determine the number of visitors expected after 9 days.

Express your answer as a whole number.

Question 4

3 marks

104

Solve algebraically:

$$10^{3x} = 7^{x+5}$$

Express your answer correct to 3 decimal places.

Question 5

2 marks

105

A word contains two Ms, two Es, two Ns, and no other repeated letters.

Suppose one of the Ns is replaced by an M.

Will this replacement result in greater or fewer permutations?

Justify your reasoning.

Question 6

3 marks

106

There is a group of 16 boys and 12 girls. How many ways can a committee of 3 people be formed if there must be at least 2 girls on the committee?

Express your answer as a whole number.

Note: A calculator is not required for the remaining test questions.

Question 7

2 marks

107

A student is using the formula $s = \theta r$ to find an arc length of a circle. Given a central angle measure of 35° and a radius of 6 cm, the student's solution is as follows:

$$s = (35)(6)$$

$$s = 210 \text{ cm}$$

Explain why this solution is incorrect.

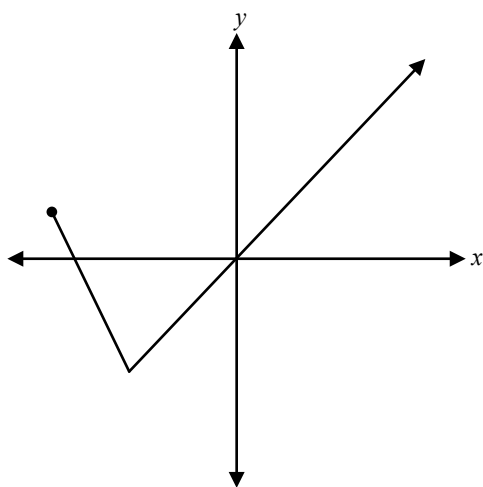
Write the correct solution.

Question 8

1 mark

108

Given the graph of $f(x)$ below, explain how you would sketch the graph of $y = |f(x)|$.



Question 9

1 mark

109

Claire correctly solves the following equation:

$$\log_2(6-x) + \log_2(3-x) = 2$$

She finds two possible values of x : $x = 2$ and $x = 7$.

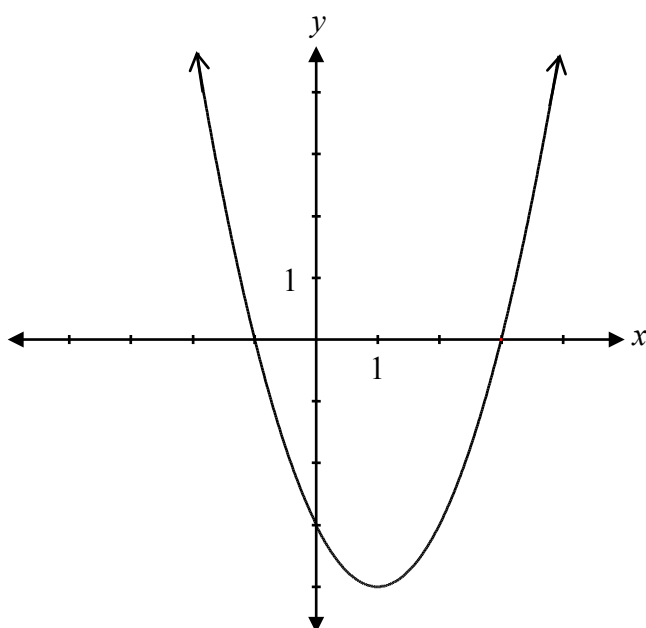
Identify which one of these values is unacceptable and explain why.

Question 10

1 mark

110

Given the graph of the function $f(x)$ below, state the domain of $y = \sqrt{f(x)}$.



Question 11

1 mark

111

A school offers 4 different Science courses, 3 different Mathematics courses, and 2 different English courses.

Julie must select 1 Science course, 1 Mathematics course, and 1 English course. She thinks this creates 9 options for her timetable.

Show why Julie is incorrect.

Question 12

1 mark

112

Explain how Pascal's triangle can be used to determine the coefficients in the binomial expansion of $(x + y)^n$.

Question 13

3 marks

113

Prove the identity below for all permissible values of x :

$$\frac{1 + \cos 2x}{\sin 2x} = \cot x$$

Left-Hand Side	Right-Hand Side

Question 14

2 marks

114

Your classmate, Leo, was absent for one of his math lessons.

Explain to Leo how to determine the cosecant ratio for an angle in standard position given that

$P(-3, -4)$ is a point on the terminal arm of the angle.

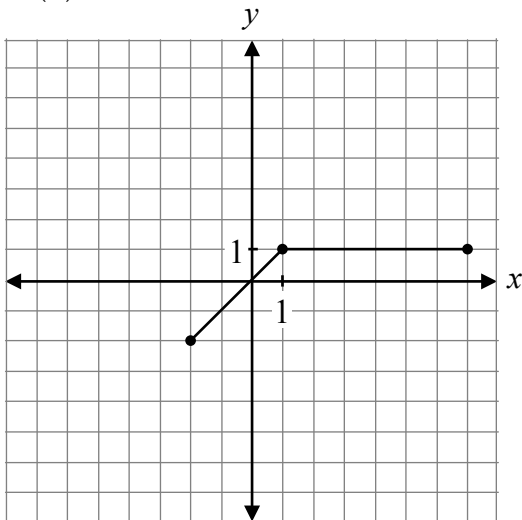
Question 15

2 marks

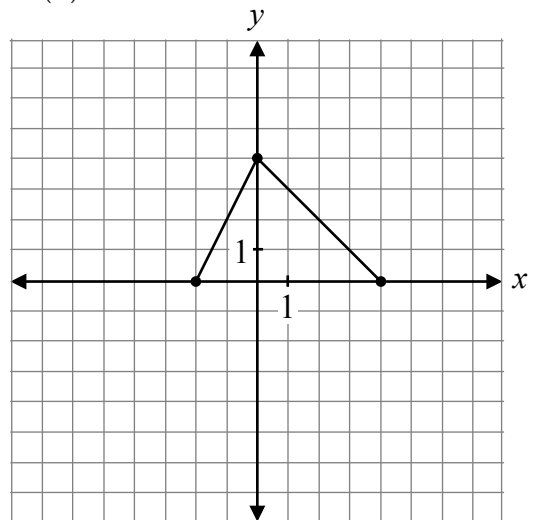
115

Given the following graphs:

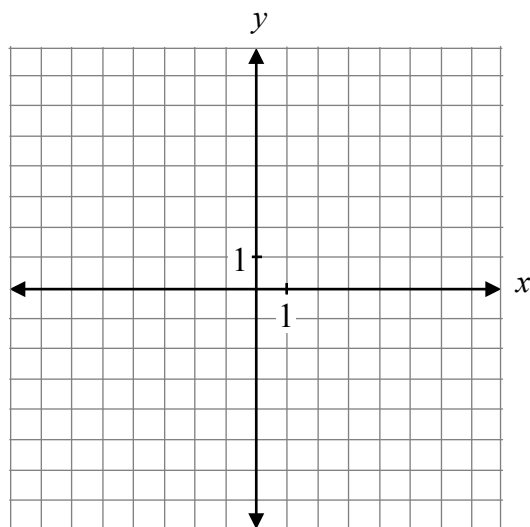
$f(x)$



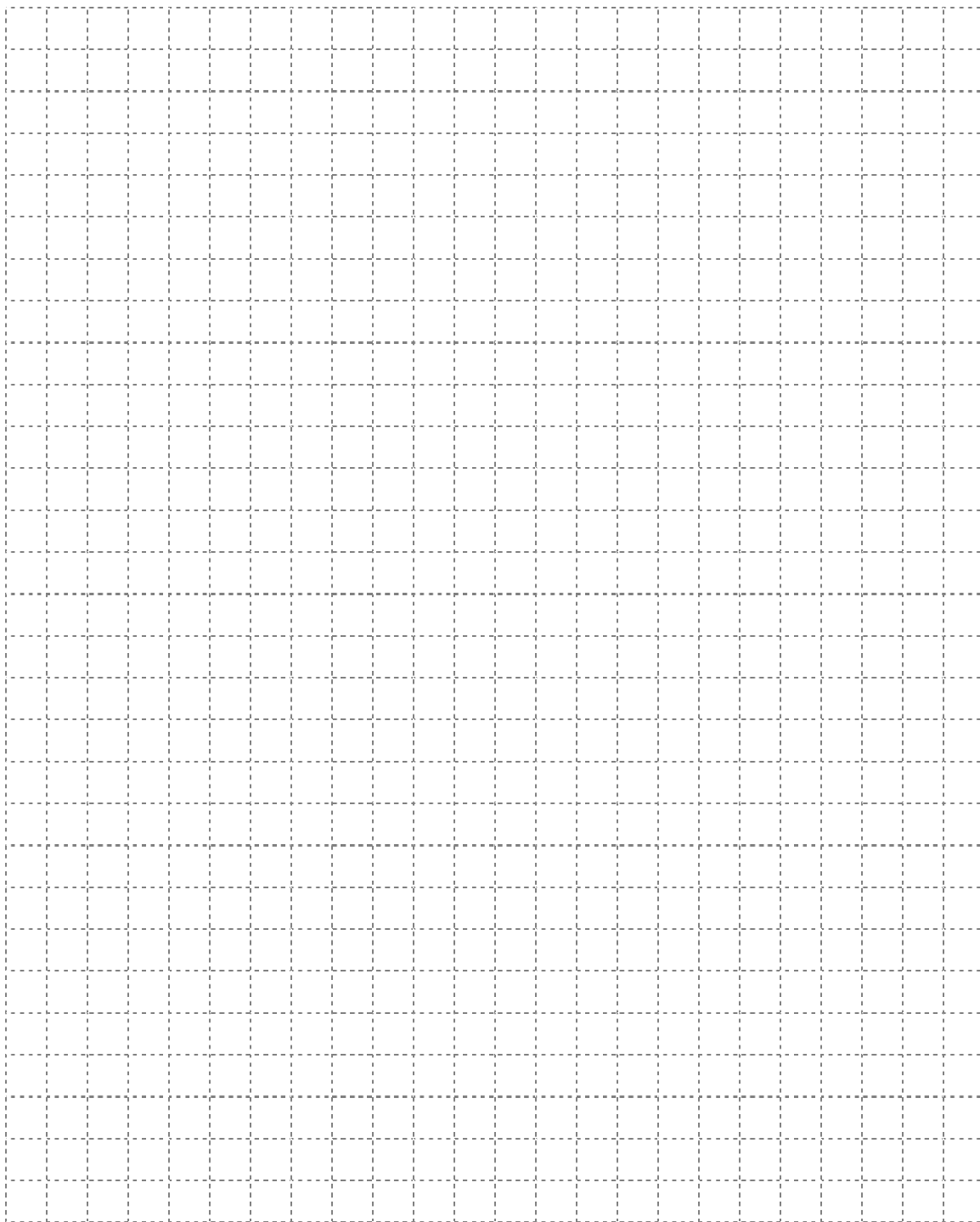
$g(x)$



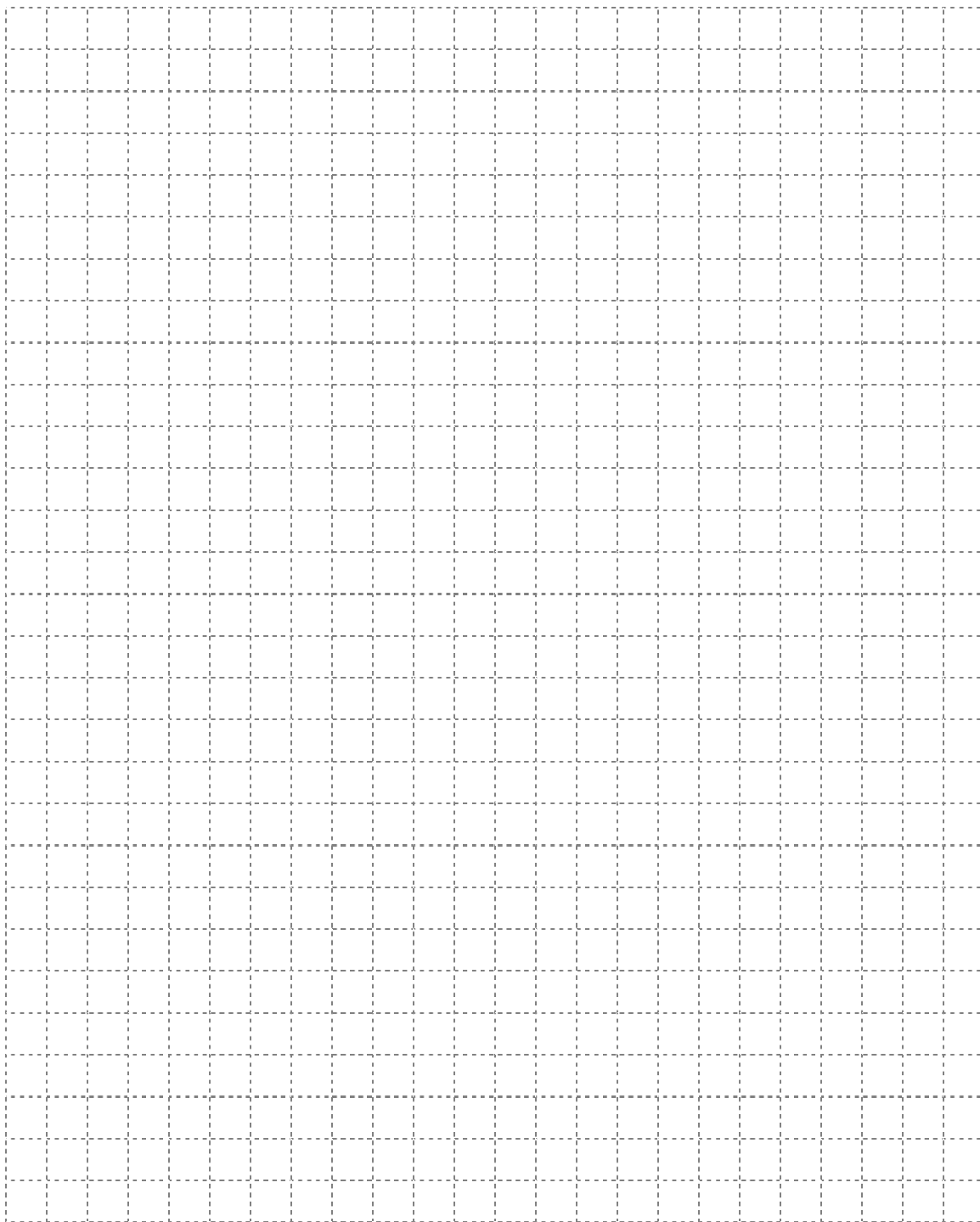
Sketch the graph of $f(x) + g(x)$.



No marks will be awarded for work done on this page.



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